

FENT2321

Wind Energy and Design of a Wind Turbine

Assignment 4: Realistic Blade Design

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1 Final Blade Design and Performance

1.1 Blade geometry

The final blade geometry obtained from the design procedure is shown in Figure 1. The corresponding blade visualisation is shown in Figure 2. For this resubmission I have manually adjusted the values for chord length for each section to comply with the constraints. These have all been adjusted by a factor of 68.33% to keep the ratios of each blade section equal to the original model.

TABLE 4			Is the chord length		Is the thickness
local radius [m]	chord length [m]	Twist [deg]	within the constraint?	thickness [m]	within the constraint?
0.03	0.0328	0.0000	Too short chord length	0.0046	Too thin
0.03	0.0861	39.5490	OK	0.0121	OK
0.0501	0.0999	31.6230	OK	0.0140	OK
0.0699	0.0986	25.4000	OK	0.0138	OK
0.09	0.0919	20.6200	OK	0.0129	OK
0.1101	0.0838	16.9410	OK	0.0117	OK
0.1299	0.0760	14.0710	OK	0.0106	OK
0.15	0.0691	11.7930	OK	0.0097	OK
0.1701	0.0630	9.9540	OK	0.0088	OK
0.1899	0.0577	8.4440	OK	0.0081	OK
0.21	0.0532	7.1860	OK	0.0074	OK
0.2301	0.0492	6.1230	OK	0.0069	OK
0.2499	0.0458	5.2160	OK	0.0064	OK
0.27	0.0428	4.4320	OK	0.0060	OK
0.2901	0.0401	3.7500	OK	0.0056	OK
0.3	0.0401	3.7500	OK	0.0056	OK

Figure 1: Visualisation of the final blade geometry.

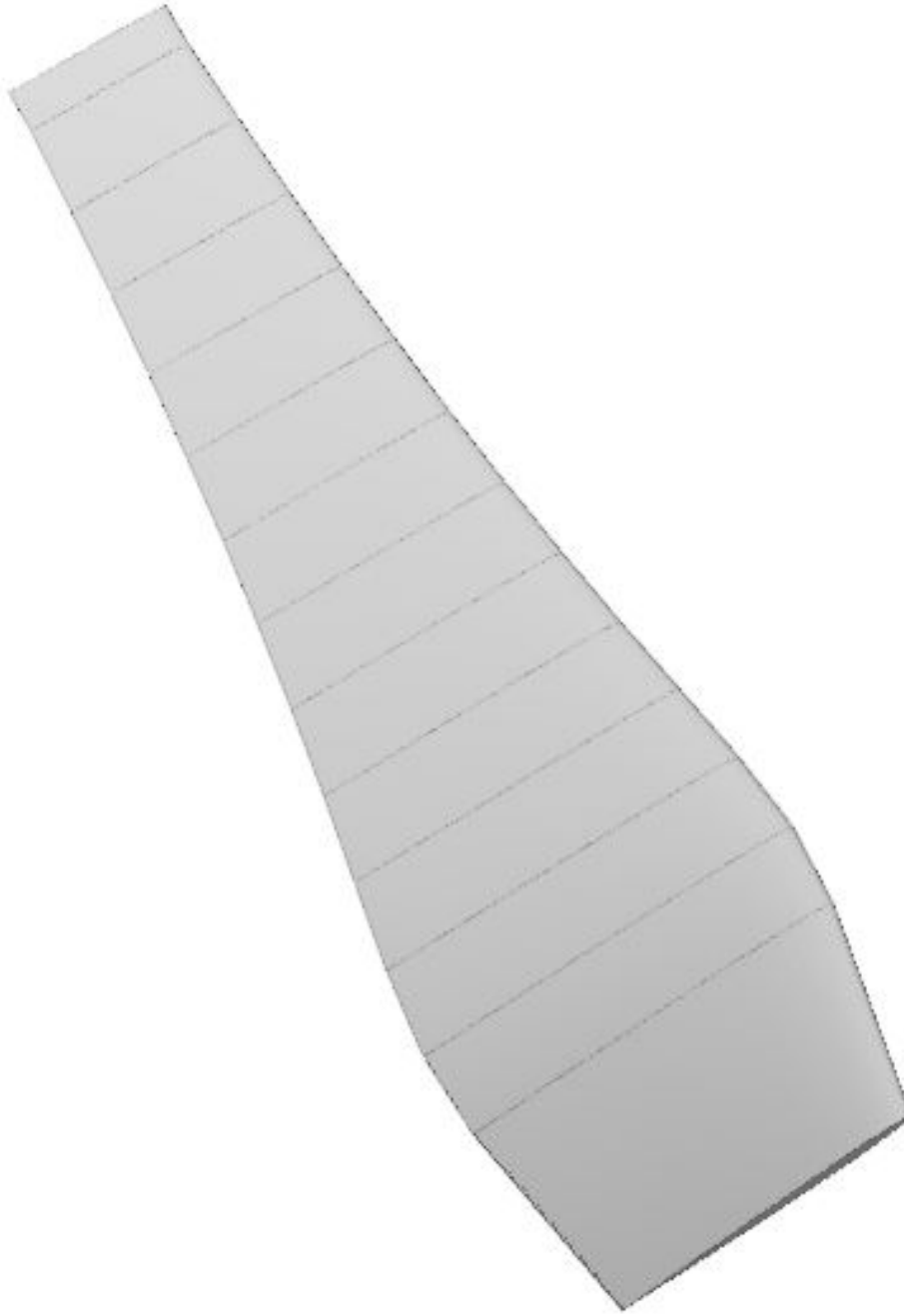


Figure 2: Visualisation of the final blade geometry.

1.2 Design-point performance

The performance of the final blade was evaluated at the design point using the simulation including Prandtl tip losses. The resulting performance quantities are summarised in Table 1. The values for C_P and C_T are without the Prandtl tip losses as this achieved the biggest values in my case. The new simulation gives better values for the parameters in the table as reasonable RPM and torque above 1Nm.

Quantity	Value
λ_{design}	6.5
$C_{P,\text{max}}$	0.345
C_T	0.842
RPM	500
Torque [Nm]	1.2

Table 1: Performance at the design point including Prandtl tip losses.

1.3 Performance curves

I tried to follow the tutorial for the C_P and C_T simulations, both with and without the Prandtl tip losses method. The red line in Figure and Figure correspond to the Prandtl method and the yellow line is the "normal" method without Prandtl. The graphs are a little better than the first submission, but still not great.

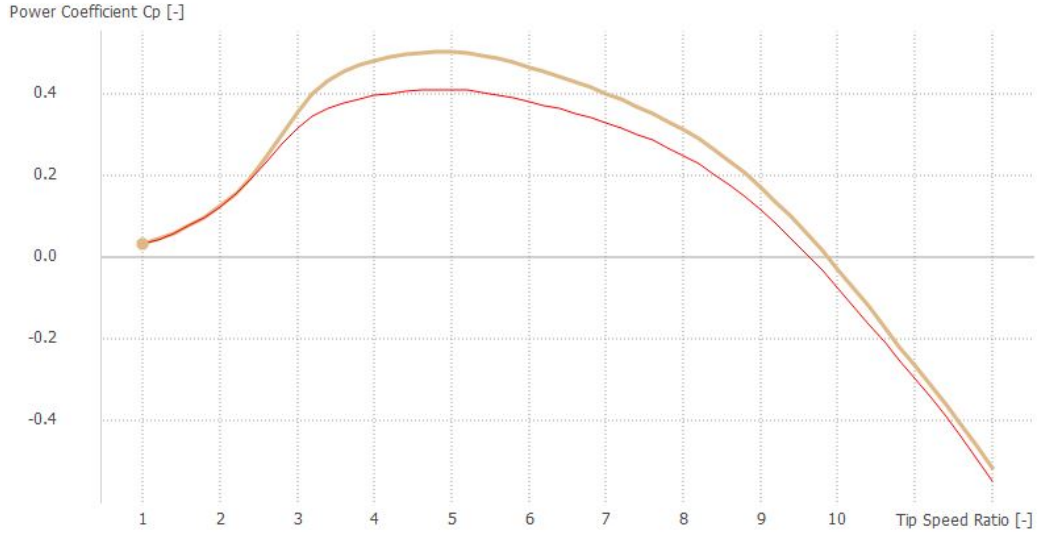


Figure 3: Power coefficient C_P as a function of tip-speed ratio λ , with and without Prandtl tip losses.

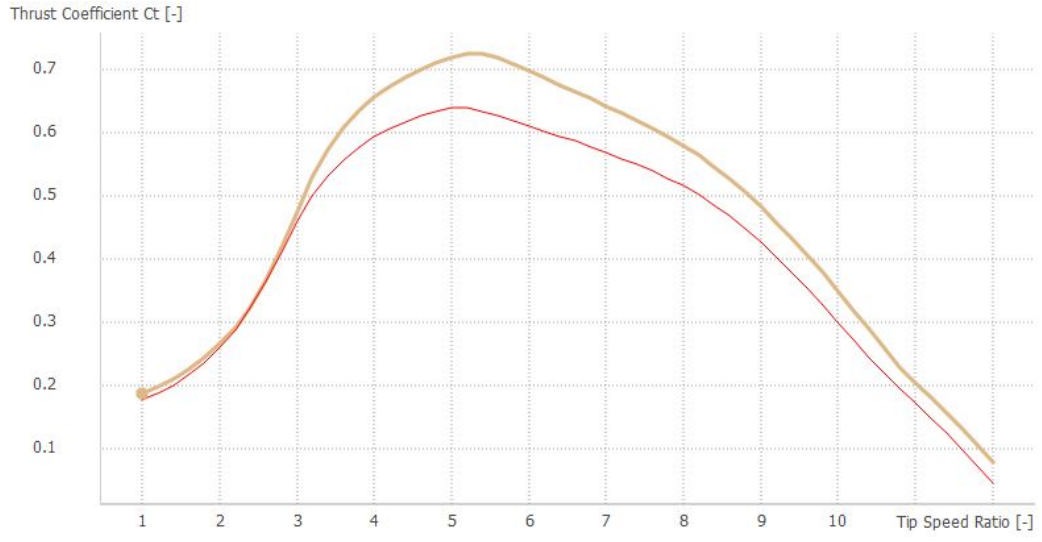


Figure 4: Thrust coefficient C_T as a function of tip-speed ratio λ , with and without Prandtl tip losses.